

■■ Computer Fundamentals

A Complete Study Guide

Covering the fundamentals of computer systems, hardware architecture, memory, expansion interfaces, and input devices — expanded with detailed explanations, real-world examples, and comparison tables for exam-ready revision.

Units Covered

Unit 1 — Introduction to Computers

Unit 2 — Internal / External Computer Cabinet Parts

Unit 3 — Input Devices

Unit 1 — Introduction to Computers

1. Basics of Computers

What is a Computer?

A computer is an electronic device that accepts raw data as input, processes it by following a set of stored instructions called a **program**, and produces meaningful, organized **information** as output. Unlike a simple calculator, a computer can store both data and instructions, make logical decisions, and repeat operations automatically without human intervention once a program has started. This ability to store and execute instructions is what separates a true computer from any other calculating device.

Characteristics of a Computer

Characteristic	Explanation
Speed	Modern computers can perform millions to billions of calculations every second, measured in units such as MHz (megahertz) or GHz (gigahertz). A task that would take a human days to calculate by hand can be completed in a fraction of a second.
Accuracy	A computer performs calculations with near-perfect accuracy as long as the input and instructions are correct. Errors that do occur are almost always due to incorrect data or flawed programming — a concept summarized as ' <i>Garbage In, Garbage Out</i> ' (GIGO).
Diligence	A computer does not suffer from fatigue, boredom, or loss of concentration. It can repeat the same task millions of times, over hours or days, with the same level of accuracy on the first attempt as on the last.
Storage Capability	Computers can store enormous volumes of data — text, images, audio, and video — permanently on devices such as hard disks and SSDs, and retrieve that data instantly whenever required.
Versatility	The same machine can be used for vastly different tasks: playing a game, preparing a payroll spreadsheet, editing a video, or browsing the web — simply by running different software.
No Feeling / No IQ (bonus)	A computer has no emotions, intuition, or judgment of its own. It cannot decide what to do on its own — it can only execute the instructions given to it by a program or a user.

Data Processing Cycle

Every computer follows the same fundamental cycle when handling information: **Data** → **Process** → **Information**. **Data** refers to raw, unorganized facts and figures (numbers, words, symbols) that by themselves carry little meaning. **Processing** is the set of operations — sorting, calculating, comparing, summarizing — the computer performs on that data according to a program's instructions. **Information** is the organized, meaningful output produced after processing, which can be used for decision-making.

Example: Entering the number of hours worked and the hourly wage rate (Data) → the computer multiplying hours by rate and applying any deductions (Process) → printing the final paycheck amount on a payslip (Information).

Additional example: A student's raw exam marks in five subjects (Data) are added and averaged by the computer (Process) to generate a report card showing the total, percentage, and grade (Information).

2. Classification of Computers by Data Processed

Computers can be grouped according to the type of data — continuous or discrete — that they are designed to process.

Analog Computers

Analog computers process continuous, fluctuating physical quantities such as voltage, temperature, pressure, or speed, rather than discrete numbers. They represent data as a continuously varying physical signal instead of the fixed binary values used by digital machines. Analog computers are generally fast for the specific physical quantity they measure but are less accurate and less flexible than digital systems, and they cannot easily store data for later retrieval.

Examples: traditional mercury or alcohol thermometers, older analog car speedometers, analog voltmeters and ammeters, and slide rules.

Digital Computers

Digital computers process data that is broken down into discrete units represented in binary form — combinations of 0s and 1s. Because information is stored as distinct, countable values rather than a continuous signal, digital computers can achieve very high accuracy, can store data permanently, and can be reprogrammed to perform an almost unlimited range of tasks. Nearly all computers in everyday use today — from smartphones to supercomputers — are digital.

Examples: laptops, desktop PCs, smartphones, tablets, digital wristwatches, and digital weighing scales.

Hybrid Computers

Hybrid computers combine the features of both analog and digital systems. They typically accept continuous analog signals from the real world, convert those signals into digital form using an analog-to-digital converter (ADC), and then process and display the data digitally for greater accuracy and easier storage. This makes hybrid computers ideal for scientific and medical instruments that must measure real-world, continuously changing quantities but present the results as precise digital readings.

Example: an ECG (Electrocardiogram) machine in a hospital continuously measures a patient's heartbeat (an analog signal) and converts it into digital data for accurate, storable, and printable readings. Petrol pumps that convert flow of fuel (analog) into a price and volume display (digital) are another common example.

Type	Data Type	Accuracy	Typical Use
Analog	Continuous signals	Lower	Measuring physical quantities
Digital	Discrete (0s & 1s)	Very high	General-purpose computing
Hybrid	Both (analog input, digital processing)	High	Scientific & medical instruments

3. Classification of Computers by Processing Capabilities

Computers are also classified by their processing power, physical size, cost, and the number of users they can support at once.

Category	Description	Example
Microcomputers	The smallest and most affordable category, designed for use by a single individual at a time. They use a microprocessor as the CPU and are what most people mean by the word 'computer' in daily life.	Desktop PCs, laptops, tablets, smartphones
Minicomputers	Mid-range, mid-sized machines more powerful than microcomputers, capable of supporting several users working simultaneously through connected terminals. Commonly used by small and medium-sized businesses for tasks such as billing and inventory management.	Small business departmental servers
Mainframe Computers	Very large, powerful, and highly reliable machines capable of supporting hundreds or thousands of users at the same time. They are used by large organizations that must process massive volumes of transactions quickly and reliably.	Banking systems processing millions of ATM/credit-card transactions worldwide, airline reservation systems
Supercomputers	The fastest, most powerful, and most expensive computers in existence, built using thousands of processors working in parallel to perform extremely complex mathematical and scientific calculations at enormous speed.	Weather forecasting, climate research, nuclear simulations, molecular and genetic modeling

4. History and Generations of Computers

The evolution of computers is commonly divided into five 'generations,' each defined by a major advance in the core electronic technology used to build them.

Generation	Period	Core Technology	Key Features / Examples
First	1940s–1950s	Vacuum tubes	Enormous in size, extremely expensive, consumed huge amounts of electricity, and generated so much heat that special cooling was required. Programmed in machine language. Example: ENIAC.
Second	1950s–1960s	Transistors	Transistors replaced vacuum tubes, making computers smaller, faster, more energy efficient, and far more reliable. Assembly language began to replace pure machine code.
Third	1960s–1970s	Integrated Circuits (ICs)	Thousands of transistors were packed onto a single silicon chip. Computers became smaller and cheaper still, and keyboards and monitors were introduced, allowing more direct human interaction.
Fourth	1970s–Present	Microprocessors	An entire CPU was placed on a single chip (the microprocessor), which brought computing power to ordinary consumers for the first time. Example: IBM PC, Apple Macintosh.
Fifth	Present & Future	Artificial Intelligence, quantum computing, parallel processing	Focused on machines that can learn, reason, and respond in natural language, alongside research into quantum and highly parallel architectures. Examples: voice assistants, large language models such as ChatGPT.

5. Simple Model of a Computer

At its core, every computer — regardless of size or purpose — follows the same basic functional model: it accepts data through **input devices**, processes that data in the **CPU**, and delivers results through **output devices**, while using **storage** to hold data and programs before, during, and after processing.

Input Devices

Hardware used to feed raw data and instructions into the computer system for processing. Example: a keyboard used to type a document, or a mouse used to click and select items on screen.

CPU (Central Processing Unit) — the 'Brain' of the Computer

The CPU is the component that actually carries out instructions and performs calculations. It is made up of three main parts working together:

- **ALU (Arithmetic & Logic Unit):** Performs all mathematical operations (addition, subtraction, multiplication, division) as well as logical/comparison operations (e.g., checking whether A is greater than B, or whether two values are equal).
- **Control Unit (CU):** Acts as the manager of the CPU. It does not process data itself but directs and coordinates the activities of all other parts of the computer — fetching instructions from memory, decoding them, and directing the ALU, memory, and input/output devices on what to do and when.
- **Internal Memory (Registers & Cache):** Extremely small, extremely fast temporary storage located directly inside the CPU, used to hold data and instructions that are being used *right now*, so the CPU does not have to wait for slower main memory.

Output Devices

Hardware used to present the processed information back to the user in a usable form. Example: a monitor displaying a document on screen, or a printer producing a paper copy.

Secondary Storage Devices

Devices used to store data and programs permanently, even after the computer is switched off (unlike the CPU's internal memory, which is temporary). Example: Hard Disk Drives (HDD), Solid State Drives (SSD), and USB flash drives.

Flow summary: Input Devices → CPU (Control Unit directs ALU and Registers) → Output Devices, with Secondary Storage available to the CPU throughout the process for reading and saving data.

Unit 2 — Internal / External Parts of the Computer Cabinet

1. The Core Components

Motherboard

The motherboard (also called the mainboard or system board) is the main printed circuit board (PCB) inside the computer cabinet. Every other major component — the CPU, RAM, storage drives, and expansion cards — connects to it either directly or through a cable, and it provides the electrical pathways (buses) that allow all these parts to communicate with one another. If the CPU is the 'brain,' the motherboard is the 'nervous system' and 'skeleton' that holds everything together.

Types of Processors (CPUs)

The processor, or CPU, has evolved from having a single processing 'core' to having multiple cores on one chip, allowing it to handle several tasks (multitasking) far more efficiently. A **core** can be thought of as an independent mini-processor inside the main chip; more cores generally mean better performance for tasks that can be split into parallel pieces, such as video editing or running many programs at once.

- **Dual Core:** A processor with two independent cores on a single chip.
- **Core 2 Duo:** An early Intel dual-core processor line, popular in the mid-to-late 2000s.
- **Intel Core i3 / i5 / i7 / i9:** Intel's modern processor tiers, ranging from entry-level (i3) to high performance (i9), each generally offering more cores, higher clock speeds, and better multitasking ability than the tier below it. *(Note: some syllabi list "i2" — this does not exist as a standard product and is most likely a typo for the i3/i5/i7/i9 family.)*

2. Memory Structure and Types

Computer memory is generally divided into two broad categories based on whether it keeps its data when the power is switched off: **volatile** memory (RAM), which loses its contents, and **non-volatile** memory (ROM), which retains its contents.

RAM (Random Access Memory)

RAM is volatile memory, meaning all of its contents are lost the instant power is removed. It acts as the computer's short-term 'working memory' — holding the operating system, open programs, and data that are currently in use so the CPU can access them far faster than it could from a hard disk. The more RAM a computer has, the more programs it can run smoothly at the same time.

- **SRAM (Static RAM):** Faster and more expensive than DRAM. It does not need to be electrically refreshed to retain its data (as long as power is supplied), which makes it ideal for very small amounts of ultra-fast storage such as CPU cache.
- **DRAM (Dynamic RAM):** Slower and cheaper than SRAM. It must be refreshed thousands of times per second to keep its stored data from fading. DRAM in its various forms (DDR3, DDR4, DDR5) is used as the main system memory in most computers.

ROM (Read-Only Memory)

ROM is non-volatile memory, meaning it keeps its data even when the computer is switched off. It is used to permanently or semi-permanently store critical startup instructions — most notably the BIOS (Basic Input/Output System) — that the computer needs the moment it is turned on, before the operating system has even loaded.

Type	Full Form	Key Feature
PROM	Programmable ROM	Can be written (programmed) only once by the manufacturer or user, using a special device; after that, it cannot be changed.
EPROM	Erasable PROM	Can be erased and reprogrammed multiple times, but erasing requires exposing the chip to strong ultraviolet (UV) light through a small window on the chip.
EEPROM	Electrically Erasable PROM	Can be erased and reprogrammed electronically, without UV light or removing the chip. This is the technology behind modern flash memory used in USB flash drives and SSDs.

RAM vs. ROM at a Glance

Feature	RAM	ROM
Volatility	Volatile (data lost on power-off)	Non-volatile (data retained on power-off)
Purpose	Temporary working memory for running programs	Stores permanent startup instructions (BIOS)
Speed	Fast read & write	Typically read-only or rarely rewritten

3. Expansion and Connections

Slots

Slots are connectors built into the motherboard that allow additional hardware components to be physically inserted and electrically connected to the system.

- **ISA Slots (Industry Standard Architecture):** An early, now obsolete expansion slot standard used in older PCs to connect basic add-on cards; largely replaced by faster standards decades ago.
- **PCI / PCIe Slots (Peripheral Component Interconnect / PCI Express):** The modern standard expansion slots used today to connect high-performance add-on cards such as graphics cards (GPUs) and dedicated sound cards. PCIe offers much greater data transfer speed than the older PCI standard.
- **Memory Slots (DIMM slots):** Slots on the motherboard specifically designed to hold RAM sticks/modules, allowing memory capacity to be increased or replaced.

Sockets

The CPU socket is the specialized connector on the motherboard into which the processor (CPU) chip is physically installed. Different CPU models require matching socket types (for example, certain Intel and AMD sockets are not interchangeable), so the socket type is a key factor when choosing a compatible processor for a given motherboard.

Cables & Ports

Port / Cable	Description
Serial	An older communication standard that transmits data one bit at a time, one after another, over a single wire pair. Slower but historically simple and reliable for devices like older modems and mice.
Parallel	An older standard that transmits multiple bits of data simultaneously over several wires at once, historically making it faster than serial for short distances. Most commonly used to connect early printers.
USB (Universal Serial Bus)	The modern, near-universal standard for connecting almost all peripherals — keyboards, mice, printers, external drives, and phones — thanks to its speed, ease of use, and 'plug and play' support (no restart required, in most cases).
PS/2	An older style of small, round, color-coded port used almost exclusively for a keyboard (purple) and a mouse (green) before USB became the standard.
HDMI (High-Definition Multimedia Interface)	A modern digital port that carries both high-definition video and audio through a single cable, commonly used to connect a computer to a monitor, projector, or television.

4. Other Vital Components

Power Devices — UPS (Uninterruptible Power Supply)

A UPS is a device containing an internal battery that sits between the electrical wall outlet and the computer. If the main power supply is suddenly cut off, the UPS instantly switches to battery power, keeping the computer running for a short period. This brief window gives the user enough time to save any open work and shut the system down safely, preventing data loss or hardware damage caused by a sudden power outage.

Expansion Cards

Card	Function
Graphic Card (GPU)	Renders and processes images, video, and 3D graphics, then sends the finished visual output to the monitor. A dedicated GPU is especially important for gaming, video editing, and design work.
Network Card (NIC — Network Interface Card)	Allows the computer to connect to a local network and/or the internet, either through a wired Ethernet cable or wirelessly via Wi-Fi.
Sound Card	Processes audio input (such as from a microphone) and audio output (such as to speakers or headphones), converting digital audio data into sound and vice versa.

Unit 3 — Input Devices

1. Introduction to Input Devices

Input devices are hardware peripherals that allow a user to interact with a computer by sending data or control signals into the system for processing. Without input devices, a computer would have no way of receiving instructions or raw data from the outside world.

2. General Types of Input Devices

Typing & Navigation Devices

- **Keyboard:** The primary device for typing text, numbers, and commands into a computer.
- **Mouse:** A pointing device used to move the on-screen cursor and select items by clicking.
- **Trackball:** Sometimes described as an 'upside-down mouse' — instead of moving the whole device, the user rolls an exposed ball with their fingers or palm to move the cursor, while the device itself stays stationary.
- **Glide-Pad (Touchpad):** The small touch-sensitive pad built into most laptops, used in place of a mouse to move the cursor and register clicks or taps.

Gaming Devices

- **Joystick:** A lever that pivots on a base and reports its angle/direction to the computer; commonly used in flight simulators and certain games.
- **Gamepad:** A handheld controller with buttons, joysticks (thumbsticks), and triggers, used to control characters and actions in video games.

Creative / Design Input Devices

- **Light Pen:** A light-sensitive, pen-shaped input device that allows a user to draw or select items directly on the screen of older CRT (Cathode Ray Tube) monitors.
- **Touch Screen:** A display surface that responds directly to the physical touch of a finger or stylus, combining an input device and an output device (the monitor) in one.
- **Digitizer & Graphic Tablet:** A flat, pressure-sensitive surface used together with a special stylus pen, allowing digital artists and designers to draw and sketch naturally, with the strokes captured directly by the computer.

Audio / Visual Input Devices

- **Microphone (Mic):** Captures sound (such as a person's voice) and converts it into digital audio data the computer can store or process.
- **Web Camera:** Captures still photos and live video, commonly used for video calls, recordings, and photography.

Commercial Input Devices

POS (Point of Sale) Terminals: Specialized input systems used at retail checkout counters (for example, at a grocery store) that combine features such as a barcode scanner, card reader, and keypad to record purchases and process payments in one integrated device.

3. Specialized Input Devices

MIDI Keyboard (Musical Instrument Digital Interface)

A MIDI keyboard looks like a piano keyboard, but rather than producing sound itself, it sends digital data — describing which notes are pressed, how hard, and for how long — to a connected computer. Music software on the computer then uses this data to generate or record sound, making MIDI keyboards a common tool in digital music production.

Wireless Input Devices

Keyboards, mice, and other input devices that connect to the computer without a physical cable, typically using Radio Frequency (RF) signals (via a small USB receiver) or Bluetooth. Wireless devices offer greater desk flexibility at the cost of needing batteries or periodic charging.

4. Types of Scanners (Data Recognition Devices)

These specialized input devices automate data entry by 'reading' specific kinds of pre-existing markings, rather than requiring a person to type the data in manually. This greatly speeds up data entry and reduces human error in high-volume settings such as banks, schools, and warehouses.

Type	Full Form	How It Works & Typical Use
OCR	Optical Character Recognition	Scans printed or typed text and converts it into an editable digital text document, allowing a paper document to be turned into an editable file on the computer.
OMR	Optical Mark Recognition	Detects the presence or absence of a mark (such as a pencil-filled bubble) at predefined positions on a form. Widely used to grade multiple-choice bubble sheets in exams and to process survey forms.
MICR	Magnetic Ink Character Recognition	Reads characters printed with special magnetic ink, most famously the string of numbers along the bottom of bank cheques. Because it relies on magnetic properties rather than just shape, MICR is highly resistant to forgery and reads reliably even if the cheque is marked or stamped over the numbers. Heavily used throughout the banking industry to process cheques quickly.
OBR	Optical Barcode Recognition	Reads the standard black-and-white zebra-striped barcodes printed on most retail products, instantly retrieving the product's price and details from a database at checkout.

End of Study Guide — Units 1 to 3. Review the comparison tables above for quick revision before exams.

■ Data Storage

A Complete Study Guide — Unit 2

Covering magnetic storage devices, how data is physically organized on a disk, the read/write process, and the key performance metrics used to measure disk speed — expanded with detailed explanations, real-world analogies, and comparison tables for exam-ready revision.

Topics Covered

1. Introduction to Data Storage
2. Types of Magnetic Storage Devices
3. Storage Mechanism of Magnetic Disks
4. Reading and Writing Data
5. Disk Performance Metrics

Unit 2 — Data Storage

1. Introduction to Data Storage

Data storage — also called **secondary memory** or **secondary storage** — refers to the hardware components that retain digital data **permanently**, even when the computer is powered off. This property is known as **non-volatile memory**, and it is what distinguishes storage devices from RAM.

RAM (primary memory) is designed for speed: it holds the operating system, open programs, and data the CPU needs *right now*, but it is volatile — everything in it disappears the instant power is lost. Storage devices, by contrast, are designed for permanence and capacity: they hold your operating system, installed software applications, documents, photos, and every other personal file for the long term, ready to be loaded into RAM whenever they are needed.

Quick distinction: RAM = fast, temporary, small capacity, volatile. Secondary Storage = slower, permanent, large capacity, non-volatile.

2. Types of Magnetic Storage Devices

Magnetic storage records data by magnetizing tiny regions of a special magnetizable material in different patterns or directions. Each magnetized region represents a binary digit — a 0 or a 1 — so a long sequence of these regions can encode any kind of digital data, from text to video.

Floppy Disk

An older, portable storage medium made of a thin, flexible ('floppy') magnetic disk sealed inside a rigid square plastic carrier for protection. Floppy disks had a very low storage capacity — the common 3.5-inch version held only about **1.44 MB**, barely enough for a few text documents by today's standards. They have been fully replaced by USB flash drives, optical discs, and cloud storage, and are now considered obsolete.

Magnetic Tape

A storage medium similar in concept to an old cassette tape, where data is recorded sequentially along the length of a long magnetic ribbon. Because data can only be accessed **sequentially** — the tape must be physically fast-forwarded or rewound to reach a particular point — magnetic tape is very slow when retrieving a specific, individual file. However, it remains extremely **cheap per gigabyte** and highly reliable over long periods, which is why it is still widely used today for massive, long-term data backups and archival storage by large organizations.

Hard Disk Drive (HDD)

The traditional primary storage device used inside most desktop and laptop computers for decades. An HDD consists of one or more rigid, rapidly spinning circular platters coated with a thin layer of magnetic material. Data is written to and read from these platters by a moving read/write head (see Section 4). HDDs offer large storage capacity at a relatively low cost per gigabyte, but because they rely on mechanical moving parts, they are slower and more fragile than newer flash-based storage.

SATA (Serial ATA)

SATA is **not a storage device itself** — it is the standard **connection cable and interface** used to connect storage drives (both HDDs and many SSDs) to the motherboard. It defines how data physically travels between the drive and the rest of the computer, and it replaced the older, bulkier IDE/PATA cable standard used in earlier PCs.

A Gentle Correction on SSDs

Many syllabi group **SSDs (Solid State Drives)** under 'magnetic storage,' but this is technically a misconception worth correcting. SSDs do **not** use magnetism or any moving mechanical parts at all — they store data electrically, in **flash memory cells**, using the same underlying EEPROM-based technology mentioned under ROM in Unit 1. Because they have no spinning platters or moving read/write arm to wait for, SSDs are significantly faster, more energy-efficient, quieter, and more physically durable (better shock resistance) than traditional magnetic hard disks — though they are typically more expensive per gigabyte of storage.

3. Storage Mechanism of Magnetic Disks

To organize the enormous number of magnetized regions on a disk so that specific data can be found quickly, a magnetic disk is physically and logically divided into well-defined regions, arranged in a hierarchy from smallest to largest.

Tracks

Tracks are concentric circles marked out on the surface of the magnetic disk, similar in appearance to the rings visible on a cut tree stump, or the grooves on a vinyl record. Each track forms a distinct circular path along which data can be written or read.

Sectors

Sectors are pie-shaped wedges that radiate out from the center of the disk, slicing across all the tracks. The small arc of a track that falls within one sector is the **smallest physical storage unit** on the disk — traditionally sized to hold **512 bytes** of data (modern drives increasingly use larger 4,096-byte / 4K sectors for greater efficiency).

Clusters

A cluster (also called an 'allocation unit') is a group of several adjacent sectors treated as a single unit. Rather than tracking and managing every individual sector one by one, the computer's operating system allocates and manages disk space in clusters, which greatly reduces the bookkeeping overhead and improves overall efficiency — at the small cost that a file smaller than one cluster still occupies the whole cluster.

Cylinders

Many hard drives contain several platters stacked on top of one another on a common spindle, each with its own read/write head. A cylinder is the vertical collection of all the tracks that line up at the exact same position across every stacked platter — imagining a single track repeated straight down through the whole stack traces out the shape of a cylinder. Because all the heads move together, the drive can access an entire cylinder's worth of tracks without needing to physically move the arm again, which is a key factor in disk performance.

Hierarchy at a Glance

Unit	Description	Analogy
Sector	Smallest physical storage unit; a wedge-shaped slice of one track (traditionally 512 bytes).	One slice of a pizza
Track	A single concentric circular path on the disk surface.	One ring on a tree stump
Cluster	A group of adjacent sectors managed together by the OS as one unit.	A bundle of pizza slices
Cylinder	The same track position aligned vertically across every platter in a multi-platter drive.	A vertical stack of matching rings across several tree-stump slices

4. Reading and Writing Data

Hard disk drives store and retrieve data using a mechanical arm tipped with a **read/write head**. Understanding how this head interacts with the spinning platters explains both how data gets onto the disk and why hard drives are comparatively slow next to electronic storage.

- The disk platters spin continuously at high speed — commonly **5,400, 7,200, or even 10,000+ RPM** (revolutions per minute) in performance drives.
- The read/write head does not physically touch the platter surface; instead, it **floats microscopically close** to it, riding on a thin cushion of air generated by the platter's own rotation.
- **To write data**, the head generates a controlled magnetic field that alters the magnetic polarity (orientation) of microscopic regions on the disk surface, encoding each region as a binary 0 or 1.
- **To read data**, the head detects the existing magnetic polarity of those same regions as they pass beneath it and translates the pattern of polarities back into digital information the CPU can use.

Because the head floats so close to the surface, any physical shock or contact between the head and the platter — known as a 'head crash' — can permanently damage the disk and destroy data, which is one of the key mechanical vulnerabilities of HDDs that SSDs do not share.

5. Disk Performance Metrics

When a computer requests a file, the hard drive cannot deliver it instantaneously — the mechanical arm and spinning platter both take a measurable amount of time to physically locate the requested data before it can even begin transferring. These delays are captured by a set of standard performance metrics.

Metric	Definition	Practical Example
Seek Time	The time it takes for the read/write mechanical arm to physically move sideways to the correct <i>track</i> on the disk.	Moving your arm to hover over the correct song groove on a vinyl record.
Rotational Delay (Latency)	The time the head must wait for the spinning platter to rotate the correct <i>sector</i> around until it is positioned directly underneath the head.	Waiting for the specific part of the song to spin back around under the record-player's needle.
Access Time	The total time required to locate the requested data before transfer can begin. Formula: Access Time = Seek Time + Rotational Delay.	The total time it takes to place the needle exactly where the song begins.
Response Time	The overall time from the exact moment the computer requests the data to the moment the data is fully delivered to RAM, ready for use — encompassing Access Time plus the actual data transfer time.	Clicking a file and waiting for it to actually open on your screen.

Remember the relationship: Seek Time + Rotational Delay = Access Time, and Access Time is just one part of the larger Response Time the user actually experiences.

End of Study Guide — Unit 2: Data Storage. Review the hierarchy and metrics tables above for quick revision before exams.

■■ Output Devices

A Complete Study Guide — Unit 3

Covering display technologies, printer types, plotters, and specialized output devices — expanded with detailed explanations, real-world examples, and comparison tables for exam-ready revision.

Topics Covered

1. Introduction to Output Devices
2. Display Units (Monitors)
3. Types of Printers
4. Plotters
5. Other Output Devices

Unit 3 — Output Devices

1. Introduction to Output Devices

An output device is any piece of computer hardware that converts the computer's internally processed digital data — ultimately just patterns of 0s and 1s — into a form a human can perceive and understand: text, graphics, tactile feedback, or audio. Output devices sit at the opposite end of the data-processing cycle from input devices: while input devices carry raw data *into* the computer, output devices carry the finished, processed information *out* of it and back to the user.

Output can be broadly split into two forms: a **soft copy**, which is a temporary, electronic version of the output visible on a screen (and which disappears once the device is switched off), and a **hard copy**, which is a permanent, physical version of the output — most commonly a printed page — that continues to exist independently of the computer.

2. Display Units (Monitors)

The monitor is the single most common visual output device connected to a computer. It creates a soft copy of the user's workspace and processed data, refreshing the image many times per second so that text, graphics, and video all appear smoothly on screen.

CRT (Cathode Ray Tube) Display Units

CRT technology is the older, bulky design used in the heavy, boxy monitors and early televisions of the 20th century. It works by firing a focused beam of electrons from an electron 'gun' at the back of a large vacuum tube toward the front of the screen, which is coated on the inside with millions of tiny phosphor dots. When the electron beam strikes these dots, they glow red, green, or blue, and the combined pattern of glowing dots forms the visible image.

CRT monitors are heavy (due to the thick glass tube needed to contain the vacuum), consume considerably more electricity than modern displays, and generate significant heat during operation. Despite these drawbacks, CRTs historically offered very fast response times and excellent color accuracy, which is why some specialized applications (such as certain gaming or broadcast setups) continued to favor them even after flat-panel alternatives became widely available.

Non-CRT Display Units (Flat-Panel Displays)

These are the modern, thin, lightweight screens found in virtually every laptop, desktop monitor, and television produced today. Unlike CRTs, they do not rely on a large vacuum tube or an electron gun, which allows them to be built at a fraction of the depth and weight.

LCD (Liquid Crystal Display)

An LCD panel uses a thin layer of liquid crystal fluid sandwiched between two sheets of polarizing glass. A fluorescent backlight positioned behind this layer shines light through it; by applying a small electrical charge, the liquid crystals can be made to twist and either block or allow that backlight to pass through each pixel, which — combined with color filters — builds up the final image.

LED (Light Emitting Diode) Display

An LED display is functionally very similar to an LCD panel — it still uses a liquid crystal layer to control light at each pixel — but it replaces the older fluorescent backlight with an array of small, energy-efficient LEDs to illuminate the screen from behind or around the edges. This change gives LED displays better contrast, thinner panels, and meaningfully lower power consumption compared to older fluorescent-backlit LCDs.

Plasma Displays

Plasma displays use tiny individual pockets of gas — commonly neon or xenon — sealed between two sheets of glass. When an electrical charge is applied to a pocket, the gas inside is converted into a glowing plasma state that emits ultraviolet light; this UV light then strikes a phosphor coating on the glass, causing it to glow with visible color, one pocket per pixel. Plasma screens were once prized for producing exceptionally deep blacks and very wide viewing angles, but they have largely been phased out in favor of LED and OLED technology due to their comparatively high power consumption, heat output, and susceptibility to permanent 'screen burn-in.'

Display Technologies at a Glance

Technology	Light Source	Notable Traits
CRT	Electron beam striking phosphor dots	Heavy, high power use, fast response, mostly obsolete
LCD	Fluorescent backlight through liquid crystals	Thin and light, moderate power use
LED	LED array through liquid crystals	Better contrast, lower power than LCD
Plasma	Glowing ionized gas pockets	Deep blacks, wide viewing angle, high power use

3. Types of Printers

Printers produce a hard copy — a permanent, physical version — of an electronic document or image. They are broadly divided into two categories based on whether the print mechanism physically strikes the paper to transfer ink.

Impact Printers

Impact printers operate by physically striking a mechanism — such as a print hammer, pin, or wheel — against an inked ribbon, which in turn presses against the paper to leave a mark, much like the mechanical action of a traditional typewriter.

***Example:** Dot Matrix Printers, which form each character out of a tiny grid (matrix) of pins striking the ribbon.*

Impact printers tend to be noisy, slower, and produce lower print quality than modern alternatives. However, because they rely on genuine physical pressure to make a mark, they are the **only** type of printer capable of printing multi-part carbonless forms in a single pass — such as carbon-copy invoices, multi-layer shipping labels, or receipt booklets — since the force of the impact is what transfers the image through every layer at once.

Non-Impact Printers

Non-impact printers form characters and images without any direct physical contact between a striking mechanism and the paper.

- **Inkjet Printers:** Spray extremely fine, microscopic droplets of liquid ink directly onto the paper surface to build up an image, one tiny dot at a time. Their ability to blend ink colors smoothly makes them especially well suited to producing high-quality color photographs.
- **Laser Printers:** Use a precisely controlled laser beam to 'draw' the shape of the document onto an electrically charged rotating drum. The charged areas of the drum then attract fine, dry toner powder, which is transferred onto the paper and permanently fused in place using heat (and pressure) from a fuser unit. Laser printers excel at high-speed, crisp, high-resolution text documents.

Overall, non-impact printers are very quiet, fast, and capable of high-resolution output, which is why they have become the standard choice for home and office printing today.

Impact vs. Non-Impact Printers

Feature	Impact Printers	Non-Impact Printers
Mechanism	Physical striking of ribbon against paper	No physical contact; ink spray or laser/toner
Noise	Loud	Quiet
Speed & Quality	Slower, lower quality	Faster, high resolution
Special Ability	Can print multi-part carbon-copy forms	Excellent for photos (inkjet) and crisp text (laser)
Examples	Dot Matrix Printer	Inkjet Printer, Laser Printer

4. Plotters

A plotter is a specialized output device designed to produce high-quality, large-scale **vector graphics** — images defined by mathematical lines, curves, and shapes rather than a grid of colored dots.

Unlike standard printers, which build up an image out of a dense grid of tiny individual dots (pixels), traditional pen plotters draw an image by physically moving one or more automated pens across the paper to trace out continuous, precise lines directly — which is what allows them to produce extremely clean, accurate line work at very large sizes without any loss of sharpness.

***Practical examples:** architects use plotters to print large building blueprints; engineers use them for CAD (Computer-Aided Design) mechanical and technical drawings; and sign-makers use specialized cutting plotters, which replace the ink pen with a small blade to cut shapes and lettering directly out of sheets of vinyl.*

5. Other Output Devices

Facsimile (FAX)

A fax machine scans a printed document, converts the scanned image into a series of audio tones, and transmits those tones over a standard telephone line to a receiving fax machine. The receiving machine decodes the tones back into an image and prints out a physical copy, effectively combining a scanner (input), a modem-like transmission process, and a printer (output) into a single device.

OLED (Organic Light Emitting Diode)

OLED is an advanced display technology in which each individual pixel is made from an organic compound that emits its own light when supplied with electrical current, rather than relying on any kind of shared backlight. Because a pixel that needs to display black can simply be turned off completely, OLED screens can achieve perfect, pitch-black darkness and extremely high contrast. This technology is widely used in high-end smartphones, premium televisions, and smartwatches.

Headphone

An audio output device worn on or inside the ears that converts the computer's electrical audio signal into sound waves the user can hear. Headphones allow a user to listen to audio privately, without the sound being broadcast to and disturbing the people around them.

SGD (Speech Generating Device)

Also known as a Voice Output Communication Aid (VOCA), an SGD is a specialized portable device that plays pre-recorded or digitally synthesized speech, typically selected by the user through buttons, a touchscreen, or even eye-tracking. These devices are a vital communication tool for individuals with severe speech impairments — such as those caused by ALS (amyotrophic lateral sclerosis) or severe cerebral palsy — allowing them to 'speak' and communicate with others despite being unable to produce speech naturally.

COM (Computer Output Microfilm)

COM is a high-speed process that transfers computer-generated data directly onto microfilm or microfiche — reels or sheets of tiny photographic film capable of storing greatly miniaturized images of documents. Because a huge number of document images can be stored on a very small physical piece of film, COM has long been used by libraries, banks, and government agencies to archive massive volumes of records compactly for long-term preservation, at far lower physical storage cost than keeping the original paper documents.

Google Glass (Optical Head-Mounted Display)

Google Glass is an example of an Optical Head-Mounted Display (OHMD) — a wearable computer output device built into a pair of glasses. It projects digital information, such as maps, notifications, or real-time translated text, directly into the wearer's field of vision, overlaying digital content on top of the real world. This blending of computer-generated output with the user's direct view of their physical surroundings is a defining example of **Augmented Reality (AR)**.

Other Output Devices at a Glance

Device	Primary Use
FAX	Transmitting scanned documents over a phone line to a remote printer
OLED	Self-lit, high-contrast displays for phones, TVs, and smartwatches
Headphone	Private audio output
SGD / VOCA	Assistive speech output for individuals with severe speech impairments
COM	Compact, long-term archival of records on microfilm/microfiche
Google Glass (OHMD)	Wearable augmented-reality display overlaying digital info on the real world

End of Study Guide — Unit 3: Output Devices. Review the comparison tables above for quick revision before exams.

■ Numbering Systems, Codes & Software

A Complete Study Guide — Unit 4

Covering binary data units, numbering systems, binary arithmetic, computer codes, parity checking, language translators, generations of programming languages, operating system types, and common software packages — expanded with detailed explanations, worked examples, and comparison tables for exam-ready revision.

Topics Covered

Part I — Numbering System and Codes

1. Binary Codes and Data Units
2. Types of Numbering Systems
3. Binary Arithmetic
4. Types of Computer Codes
5. Parity Check

Part II — Languages, Operating Systems & Software

1. Translators
2. Types of Languages
3. Types of Operating Systems
4. Uses and Applications of Software Packages

Unit 4 — Numbering System and Codes

1. Introduction to Binary Codes and Data Units

At the most fundamental level, computers are built from electronic switches (transistors) that can only be in one of two physical states: ON or OFF. We represent this two-state reality mathematically using the **binary system**, a number system with only two digits, 0 and 1 — typically 0 represents OFF and 1 represents ON. Every piece of data a computer stores or processes, no matter how complex, is ultimately built up from combinations of these two digits.

Basic Units of Data

Unit	Size	Description
Bit (Binary Digit)	1 bit	The smallest possible unit of data in a computer, representing a single 0 or 1.
Nibble	4 bits	A group of 4 bits — exactly enough to represent a single hexadecimal digit (0-F).
Byte	8 bits	A group of 8 bits, and the standard unit used to store a single character, such as the letter 'A'. Most modern storage and memory capacities are measured in bytes and their multiples.

Special Purpose Bits

- **Carry Bit:** Used during binary arithmetic whenever an addition in one column produces a result too large to fit in that column (i.e., the result is 2 or more); the excess amount is 'carried over' and added into the next column to the left, exactly as with carrying in ordinary decimal addition.
- **Sign Bit:** The leftmost (most significant) bit in a signed binary number, used to indicate whether the overall value is positive or negative. By convention, a sign bit of 0 means the number is positive, and a sign bit of 1 means the number is negative.
- **Parity Bit:** An extra bit appended to a binary message purely for error-checking purposes, allowing the receiving system to detect whether the data was corrupted in transit. This is explained in full detail in Section 5.

Storage Multipliers

As data accumulates, it is measured in progressively larger units, each one built from 1,024 (2^{10}) of the unit below it.

Unit	Size
KB (Kilobyte)	1,024 Bytes
MB (Megabyte)	1,024 KB
GB (Gigabyte)	1,024 MB

Unit	Size
TB (Terabyte)	1,024 GB
PB (Petabyte)	1,024 TB
EB (Exabyte)	1,024 PB

A Gentle Correction on Storage Units

If your syllabus lists an abbreviation such as 'HB' after TB, this is almost certainly a typo. Following Terabyte (TB), the next standard units in the sequence are **PB (Petabyte)** and **EB (Exabyte)**, each equal to 1,024 times the unit before it — there is no standard data unit abbreviated 'HB'.

2. Types of Numbering Systems

A numbering system defines the set of digits (or symbols) available and the place value each position represents. Computers work natively in binary, but several other numbering systems are commonly used because they map onto binary in a clean, convenient way.

System	Base	Digits Used	Example
Binary	Base-2	0, 1	1011_2
Octal	Base-8	0 – 7	75_8
Decimal	Base-10	0 – 9	129_{10}
Hexadecimal	Base-16	0 – 9, A – F (A=10 ... F=15)	$1A3F_{16}$

Decimal (Base-10) is the standard everyday counting system used by humans, built around the ten digits 0 through 9 — most likely because humans have ten fingers. **Binary (Base-2)** is the native language of computer hardware, since each digit maps directly onto one electronic switch being ON or OFF. **Octal (Base-8)** and **Hexadecimal (Base-16)** are used as convenient shorthand for binary: because $8 = 2^3$ and $16 = 2^4$, each octal digit corresponds exactly to 3 binary bits and each hexadecimal digit corresponds exactly to 4 binary bits (one nibble), making long binary strings far easier for humans to read, write, and debug — for example, in memory addresses and color codes.

3. Binary Arithmetic

Addition

Binary addition follows the same logical process as decimal addition, but with only two digits to work with, so any result of 2 or more must be carried:

Basic Binary Addition Rules

```

0 + 0 = 0
0 + 1 = 1
1 + 0 = 1
1 + 1 = 10 (i.e., 0, carry 1 to the next column)

```

Subtraction Using Complements

Computer hardware is built to add very efficiently, but subtracting directly would require separate, more complex circuitry. Instead, computers perform subtraction by converting the number being subtracted into its negative equivalent and then simply **adding** it — a technique made possible using complements.

- **1's Complement:** Found by simply flipping every bit in the number — every 0 becomes a 1, and every 1 becomes a 0.
- **2's Complement:** Found by first taking the 1's complement, then adding 1 to the result. This is the standard method computers actually use to represent negative numbers and perform subtraction, because — unlike 1's complement — it avoids the awkward problem of having two different representations for zero.

Worked Example: Finding the 2's Complement of 1010

```

Original number: 1010
Step 1 - Flip every bit (1's complement):
0101
Step 2 - Add 1 to get the 2's complement:
0101
+ 1
-----
0110

```

Multiplication & Division

Rather than using long multiplication or division as in decimal arithmetic, binary multiplication and division can be performed efficiently by **shifting bit positions**. Shifting every bit in a binary number one position to the **left** effectively multiplies the number by 2, while shifting every bit one position to the **right** effectively divides the number by 2. Repeated shifting combined with addition (for multiplication) or subtraction (for division) allows a computer's ALU to carry out full multiplication and division operations using very simple, fast underlying hardware.

4. Types of Computer Codes

A computer code is a standardized scheme for representing human-readable characters — letters, digits, punctuation, and symbols — as specific patterns of binary bits, so that text can be consistently stored, transmitted, and displayed by different computer systems.

Code	Bit Size	Description
ASCII	7-bit or 8-bit	American Standard Code for Information Interchange. A widely used code in older PCs that represents the standard English alphabet, digits, punctuation, and basic control characters; a 7-bit ASCII code can represent 128 distinct characters, while extended 8-bit versions can represent 256.
BCD	4 bits per digit	Binary Coded Decimal. Rather than converting an entire decimal number into one binary value, BCD encodes <i>each individual decimal digit</i> separately using its own 4-bit binary sequence — for example, the decimal number 29 would be stored as the two 4-bit groups for '2' and '9' rather than as a single binary conversion of 29.
EBCDIC	8-bit	Extended Binary Coded Decimal Interchange Code. An 8-bit character encoding developed by and primarily used on older IBM mainframe computers, largely independent of the ASCII standard used elsewhere.
Unicode	Up to 32-bit	The modern, universal character-encoding standard. Because it can use up to 32 bits per character, Unicode can represent an enormous range of characters — covering virtually every written language in the world, mathematical symbols, and even emojis — making it the standard for modern operating systems, programming languages, and the web.

5. Parity Check

A parity check is a simple, lightweight method used to detect errors that may occur when data is transmitted over a network or stored in memory. Before sending a byte of data, an extra bit — called the **parity bit** — is added to it. The value of this parity bit (0 or 1) is chosen according to a rule agreed upon in advance by both the sending and receiving systems, so that the receiver can check whether the data arrived correctly.

Even Parity System

The parity bit is set to whichever value (0 or 1) makes the **total** number of 1s in the complete data string (original bits plus the parity bit) an **even** number.

Odd Parity System

The parity bit is set to whichever value makes the total number of 1s in the complete data string an **odd** number.

Once the data reaches its destination, the receiving system counts the number of 1s in the string it received and compares this against the parity rule it was expecting. If a computer configured to expect **odd** parity instead receives a string containing an **even** number of 1s, it immediately knows that at least one bit was flipped somewhere during transmission, and it can flag the data as corrupted — for example, by requesting that the data be resent.

Worked Example: Adding a Parity Bit

Original 7-bit data: 1 0 1 1 0 0 1 (contains four 1s)

Even Parity: four 1s is already even, so parity bit = 0

Transmitted byte: 1 0 1 1 0 0 1 0

Odd Parity: four 1s is even, so we need one more 1 -> parity bit = 1

Transmitted byte: 1 0 1 1 0 0 1 1

Note: a simple parity check can reliably detect a single flipped bit, but it cannot detect an even number of flipped bits (e.g., exactly two errors can cancel each other out and go unnoticed), which is why more advanced error-detection and error-correction schemes are used in critical systems.

Unit 4 (Part II) — Languages, Operating Systems & Software

1. Translators

A computer's hardware can only directly execute instructions written in pure binary **machine language**. Since writing entire programs directly in binary is impractical for humans, programmers write code in more human-readable programming languages, and special programs called **translators** convert that human-readable code into the binary machine code the CPU can actually execute.

Translator	What It Translates	Key Characteristics
Assembler	Low-level Assembly Language → Machine Code	Converts short mnemonic instructions (such as ADD or MOV) into their exact binary machine-code equivalents, on a largely one-to-one basis.
Compiler	High-level language (e.g., C, Java) → Machine Code	Reads and translates the <i>entire</i> program at once, producing a standalone executable file ahead of time. This upfront translation step takes extra time before the program can be run, but the resulting compiled program then runs very fast, since no further translation is needed at runtime.
Interpreter	High-level language (e.g., Python, PHP) → Executed directly	Translates and executes the program <i>line by line</i> , without producing a separate standalone executable file first. This makes it easier to test and debug code interactively, but the program generally runs slower than an equivalent compiled program, since translation happens repeatedly while the program is running.

2. Types of Languages

Programming languages are commonly classified into 'generations,' reflecting how far removed each is from the computer's native binary machine code and how much abstraction and convenience each offers to the programmer.

Generation	Description	Example
1GL — Machine Level Language	Pure binary (0s and 1s), the only form of code the CPU can execute directly. Extremely difficult and error-prone for humans to read or write.	Raw binary instructions
2GL — Assembly Level Language	Replaces raw binary with short, human-readable mnemonics such as ADD, SUB, and MOV. Still closely tied to specific hardware, and requires an Assembler to convert it into machine code.	Assembly language
3GL — High-Level Language	English-like languages that are largely independent of the underlying computer hardware, making programs easier to write, read, and port between different machines.	C, Java, PHP
4GL — Very High-Level Language	Designed for specific, focused tasks rather than general-purpose programming — most notably database management and data querying, letting users specify <i>what</i> data they want without describing <i>how</i> to retrieve it step by step.	SQL
5GL — Fifth-Generation Language	Focused on Artificial Intelligence and logic programming. The programmer describes the problem and any known constraints, and the language/system itself works out the logical steps needed to reach a solution.	Prolog and similar logic-programming languages

3. Types of Operating Systems

The Operating System (OS) is the master control program that manages a computer's hardware and software resources, acting as the intermediary between the user, application programs, and the underlying hardware. Different types of operating systems are designed around different goals — throughput, responsiveness, or strict timing guarantees.

Batch Operating System

In a batch OS, users do not interact directly with the running system. Instead, similar jobs are collected together into groups called 'batches' by an operator, and the whole batch is then processed sequentially by the computer without further user input, one job after another. This approach maximizes efficient use of the computer's processing time for repetitive, non-interactive workloads, such as payroll processing.

Multi-Processing Operating System

A multi-processing OS is designed to take advantage of a computer containing **two or more physical CPUs** within the same system, allowing it to genuinely execute multiple programs at the exact same time — one (or more) per available processor — rather than merely appearing to do so.

Time-Sharing Operating System

A time-sharing OS allows many different users to share a single computer system by switching the CPU's attention between them extremely rapidly, giving each user a very short 'time slice' of processing time in turn. Because these switches happen so quickly, each individual user experiences the illusion of having dedicated, uninterrupted use of the entire machine, even though the underlying CPU is actually being divided among many users.

Online and Real-Time Operating System (RTOS)

An RTOS is built for tasks with strict, non-negotiable time constraints, where a computation or response must be completed within a guaranteed, often extremely short window — sometimes measured in milliseconds or less — or the result is considered a failure, regardless of whether the answer is otherwise correct.

***Examples:** air traffic control systems, which must process aircraft position data and issue guidance without delay, are a classic textbook example of RTOS use. Financial systems that run automated, algorithmic trading strategies and maintain precise timestamped execution logs for orders placed just before a market opens are another real-world example, since even millisecond-level timing differences can materially affect the outcome of a trade.*

***Note:** RTOS platforms carry out extremely time-sensitive automated actions, so real systems of this kind are built, tested, and regulated with great care by specialists; this study guide describes RTOS only at a conceptual level for exam purposes and is not guidance for building or operating such a system.*

Operating System Types at a Glance

OS Type	Core Idea	Typical Use
Batch	Jobs grouped and processed sequentially, no direct user interaction	Payroll runs, large repetitive back-office jobs
Multi-Processing	Two or more CPUs run programs truly simultaneously	Servers, high-performance workstations
Time-Sharing	CPU rapidly switches between many users' tasks	Multi-user mainframes, shared computing systems
Real-Time (RTOS)	Guarantees a response within a strict, short time limit	Air traffic control, industrial control systems, algorithmic trading

4. Uses and Applications of Software Packages

Beyond the operating system itself, most day-to-day computer use happens through **application software** — programs designed to help a user accomplish a specific type of task.

Category	Purpose	Example Software
Word Processing	Creating, editing, and formatting text-based documents such as letters, reports, and essays.	MS Word
Spreadsheet	Organizing data into rows and columns, performing calculations and formulas, and carrying out analytical tasks such as technical backtesting of numerical strategies.	MS Excel
Graphical	Creating and editing illustrations, photographs, and other visual designs.	Adobe Photoshop
Database	Storing, organizing, and efficiently retrieving very large amounts of structured data.	MySQL, MongoDB
Presentation	Creating visual slideshows for lectures, pitches, or meetings.	MS PowerPoint
Animation / Video / Sound	Multimedia packages used for editing audio tracks, rendering 3D animations, or compiling and editing video sequences.	Adobe Premiere

End of Study Guide — Unit 4: Numbering System and Codes, and Part II: Languages, Operating Systems, and Software. Review the comparison tables and worked examples above for quick revision before exams.

■ Emerging Technologies & Computer Virus

A Complete Study Guide — Unit 5

Covering geolocation and communication technologies, communication devices, computer viruses and malware, cloud computing, and a full glossary of key computing terms — expanded with detailed explanations, real-world examples, and comparison tables for exam-ready revision.

Topics Covered

1. Different Communication Methods
2. Communication Devices
3. Virus (Computer Malicious Software)
4. Cloud Computing
5. Important Terms and Acronyms Glossary

Unit 5 — Emerging Technologies and Virus

1. Different Communication Methods

GIS (Geographic Information System)

A GIS is a computer system designed to capture, store, manipulate, analyze, manage, and present all types of geographical spatial data — that is, data tied to a specific location on Earth. By layering different types of spatial data (such as roads, terrain, population density, or utility lines) on top of one another, a GIS lets analysts spot patterns and relationships that would be difficult to see in a plain list of numbers. It is heavily used in urban planning (deciding where to build new infrastructure), logistics (optimizing delivery routes), and environmental tracking (monitoring deforestation or pollution over time).

GPS (Global Positioning System)

GPS is a satellite-based radionavigation system owned and operated by the US government. A network of orbiting satellites continuously broadcasts precise timing and position signals; a GPS receiver on the ground calculates its own exact location by comparing the signals received from several satellites at once. This allows GPS to provide precise geolocation and time information to a receiver essentially anywhere on Earth with a clear view of the sky.

CDMA & GSM

CDMA (Code Division Multiple Access) and GSM (Global System for Mobile communications) were the two major competing radio standards used by older-generation cell phone networks.

- **GSM** became the dominant global standard and identified a subscriber to the network using a removable SIM (Subscriber Identity Module) card, which meant a user's identity and account details could easily be moved between different phones simply by moving the SIM card.
- **CDMA** was used primarily in the United States and linked a subscriber to the network by registering the specific phone hardware itself, generally without any removable SIM card.

Modern 4G and 5G networks are built on newer, unified technical standards and have largely superseded both older CDMA and GSM systems.

2. Communication Devices

Cell Phones

Portable devices used for voice and data communication, connecting to a network of fixed ground-based base stations known as 'cell sites' — each covering a limited geographic area, or 'cell' — which allows the network to seamlessly hand a call or data connection off from one site to the next as the user moves around.

Modem (Modulator-Demodulator)

A modem is a hardware device that converts the digital data used inside a computer into analog signals suitable for travelling over telephone lines (a process called **modulation**), and converts incoming analog signals back into digital data the computer can use (a process called **demodulation**) — hence the name 'MOdulator-DEModulator.'

Infrared

Infrared communication uses invisible pulses of light, just outside the range of human vision, to transmit data over short, direct line-of-sight distances. Because the receiving device must have a clear, unobstructed line of sight to the transmitter, infrared is best suited to simple, close-range control tasks. **Example:** a television remote control sending channel and volume commands to a TV.

Bluetooth

Bluetooth is a wireless technology standard for exchanging data between fixed and mobile devices over short distances, using short-wavelength UHF (ultra-high frequency) radio waves rather than light. Unlike infrared, Bluetooth does not require a direct line of sight, which is why it is widely used for connecting wireless headphones, keyboards, mice, and other nearby accessories.

WiFi (Wireless Fidelity)

WiFi refers to a family of wireless networking technologies that allow devices such as laptops, phones, and smart appliances to connect to the internet or a local network through a wireless router, without needing a physical network cable.

LiFi (Light Fidelity)

LiFi is a newer, high-speed wireless communication technology that transmits data by rapidly flickering pulses of light from ordinary LED bulbs, rather than using radio waves as WiFi does. Because it operates on the light spectrum rather than the increasingly crowded radio spectrum, LiFi can offer much higher potential speeds and greater security (light does not pass through solid walls, which naturally contains the signal to a room) than WiFi — but it requires a clear line of sight between the LED transmitter and the receiver, since light cannot bend around obstacles the way radio waves often can.

SLM (Spatial Light Modulator)

A Spatial Light Modulator is an advanced device used in optical computing and in modern projectors to precisely modulate — that is, adjust and control — the amplitude, phase, or polarization of light waves across space and over time. By controlling light this precisely, SLMs make it possible to project sharp images and are also being explored as building blocks for future optical computing systems, which aim to process data using light rather than electricity.

Wireless Communication Technologies at a Glance

Technology	Medium	Line-of-Sight Needed?	Typical Range
Infrared	Light pulses	Yes	Very short (a few meters)
Bluetooth	UHF radio waves	No	Short (up to ~10m typical)
WiFi	Radio waves	No	Medium (building/home range)
LiFi	Visible light pulses	Yes	Room-scale, very high speed

3. Virus (Computer Malicious Software)

Introduction

A computer virus is a type of malicious software (malware) that, once executed, replicates itself by modifying other legitimate computer programs and inserting a copy of its own code into them — much like a biological virus needs a host cell to reproduce, a computer virus needs a host program or file to spread.

Origin and History

- **Creeper (1971):** Often considered the first computer virus, created on ARPANET (the predecessor to today's internet). It did not cause damage but simply displayed the playful message, 'I'm the creeper, catch me if you can!' as it moved between connected computers.
- **Elk Cloner (1982):** Widely regarded as the first virus to spread 'in the wild' outside of a single research or computer lab. It targeted Apple II computers and spread by copying itself onto floppy disks, infecting any other Apple II machine that later used one of those disks.

Types of Virus

Type	How It Spreads / Behaves
Boot Sector Virus	Infects the master boot record of a hard drive — the very first section of code the computer reads when starting up — allowing the virus to load into memory before the operating system itself has even started.
File Infector	Attaches itself to executable program files, such as those ending in .exe or .com, and activates whenever the infected program is run.
Macro Virus	Written in the built-in macro (automation) scripting language of applications such as Microsoft Word or Excel, and triggers automatically when an infected document is opened.
Worm	A standalone piece of malware that can replicate itself and spread to other computers across a network entirely on its own, without needing to attach itself to any host file or program — unlike a true virus.
Trojan Horse	Malicious code disguised as legitimate, useful, or desirable software, tricking the user into voluntarily installing it themselves — named after the Trojan Horse of Greek mythology.

Problems Caused by Viruses

Viruses and other malware can cause a wide range of harm, including corrupting or destroying stored data, causing programs or the entire system to crash or behave unpredictably, and quietly stealing computing resources (such as processing power, memory, or network bandwidth) — for example, to mine cryptocurrency or send spam without the owner's knowledge.

Protection

- Installing and regularly updating reputable **antivirus software**, which scans for and removes known malicious code.
- Keeping the **operating system and all installed software updated**, since updates often patch security vulnerabilities that malware is designed to exploit.
- Using **hardware and/or software firewalls** to monitor and control incoming and outgoing network traffic.

- Exercising caution with, and generally **avoiding suspicious email attachments or downloads** from untrusted sources, since these remain one of the most common ways malware initially infects a system.

4. Cloud Computing

Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, and analytics — over the internet ('the cloud'), rather than relying on a local hard drive or a private, on-site server that an individual or organization must buy, house, and maintain themselves.

Service Models

Model	What It Provides	Example
IaaS (Infrastructure as a Service)	Renting basic IT infrastructure — servers, virtual machines, and storage — from a cloud provider, typically billed on a pay-as-you-go basis. The customer still manages the operating system and everything running on top of it.	AWS EC2, Google Compute Engine
PaaS (Platform as a Service)	Provides a ready-made framework and environment for developers to build, test, and deploy applications, without needing to worry about managing the underlying servers, operating systems, or infrastructure.	Deploying a full-stack Node.js and React application on a platform such as Heroku or Vercel
SaaS (Software as a Service)	Delivers a complete, ready-to-use software application over the internet on demand, typically on a subscription basis, with the provider handling all underlying infrastructure and maintenance.	Google Workspace, Microsoft 365, Zoom

Architecture

A cloud computing system is generally divided into two conceptual halves: the **Front-End**, which is the client's own computer along with the application (such as a web browser or app) required to access the cloud; and the **Back-End**, which consists of the massive servers, data storage systems, and computers, typically housed in large data centers, that actually make up 'the cloud' itself and do the real work of storing and processing data.

Security & Privacy

Because cloud providers store data belonging to enormous numbers of customers, they implement strict security measures to protect it, including **data encryption** (scrambling data so it cannot be read without the correct key, applied both while data is stored — 'at rest' — and while it is being transmitted — 'in transit'), **IAM (Identity and Access Management) policies** (controlling precisely who is allowed to access which resources), and **automated threat detection** systems that continuously monitor for suspicious activity.

5. Important Terms and Acronyms Glossary

A quick-reference glossary of additional key computing terms and acronyms, expanded with fuller explanations for each entry.

Term	Definition
ATM (Asynchronous Transfer Mode)	A telecommunications standard for digital transmission that can carry multiple types of traffic — including telephony, data, and video signals — together over a single network, without needing separate dedicated networks for each type of traffic.
Backup / Restore	Backup is the process of making copies of data so that these additional copies can later be used to recover the original after a data-loss event (such as accidental deletion, hardware failure, or a malware attack). Restore is the process of retrieving that previously saved backup data and putting it back into active use.
Hard Copy / Soft Copy	A hard copy is a physical, printed version of a document that exists independently of any computer. A soft copy is an unprinted, purely digital version of a document, viewed on a screen.
Bus / Data Bus	A bus is an internal communication pathway that transfers data between components inside a computer. The Data Bus specifically carries the actual data being processed, while related buses — the Address Bus and Control Bus — carry memory addresses and control/timing signals respectively.
Buffer and Spooling	A Buffer is a temporary memory area used to hold data briefly while it is being moved from one place to another, smoothing out differences in speed between the source and destination (for example, video buffering while streaming). Spooling (Simultaneous Peripheral Operations On-Line) is a specific application of buffering in which data is held in a queue while waiting for a slower peripheral device — most commonly a printer — to become ready to process it.
Cursor / Pointer / Icon	The Cursor is the blinking text marker that shows exactly where newly typed text will appear. The Pointer is the on-screen arrow (or similar shape) controlled by a mouse or trackpad, used to select and interact with items. An Icon is a small graphical image representing a program, file, or function that can be clicked to open or run it.
E-Mail / Attachment	Electronic mail (e-mail) is a message sent electronically over a network. An attachment is a separate computer file — such as an image or document — sent along with the main text of an email message.
CLI / GUI	A CLI (Command Line Interface) is a text-based interface in which the user interacts with the operating system by typing precise manual commands, as in MS-DOS or a Linux terminal. A GUI (Graphical User Interface) is a visual interface that lets the user interact using graphical elements such as windows, icons, and menus, as in Windows 11 or macOS.
Compiler and Its Types	A compiler is a program that translates high-level source code into machine code. A Cross-Compiler runs on one type of platform but produces machine code intended to run on a different platform (for example, compiling code on a desktop computer for a smartphone). A Native Compiler produces machine code intended to run on the very same type of platform it is running on.

Term	Definition
Drive / Directory (Folder) / File / Path	A Drive is a physical or logical storage device, such as the C: Drive. A Directory (or folder) is used to organize related files together. A File is a self-contained piece of information, such as a document, image, or program. A Path is the exact string of characters describing precisely where a file is located, for example C:\Users\Documents\report.txt.
Menu / Popup Menu / Toolbar	A Menu is a list of commands or options presented to the user, usually along the top of an application window. A Popup (or Context) Menu is a hidden menu that only appears in response to a specific user action, typically a right-click. A Toolbar is a row of clickable icons representing frequently used tools or commands for quick access.
Shutdown / Reboot / Restart	Shutdown completely powers the computer off. Reboot (or Restart) powers the operating system down and then immediately starts it back up again in the same action, which clears the contents of RAM in the process — often used to resolve minor software glitches.
Syntax / Wild Card Characters	Syntax refers to the strict spelling, punctuation, and grammar rules that source code in a given programming language must follow to be valid. Wildcard characters are special symbols (such as * or ?) used in searches to stand in for one or more other characters — for example, searching *.jpg will find every file ending in the .jpg extension, regardless of its name.
Optical Fiber (Fiber Optic)	A flexible, transparent fiber made of high-quality glass or plastic that transmits data as pulses of light rather than electrical signals, enabling incredibly fast internet and data transmission speeds over very long distances with minimal signal loss.
Net Meeting	Refers either to older dedicated software of that name or, more generally, to the broad concept of conducting online virtual meetings that include video, audio, and screen-sharing between participants in different locations.
Peripherals	Any auxiliary device that connects to a computer to put information in or get information out, without being part of the computer's core processing hardware (the CPU, motherboard, and internal memory). Peripherals include both input devices, such as a mouse and keyboard, and output devices, such as a monitor and printer.

Printing Speed Metrics

Different categories of printers are conventionally measured using different speed metrics, reflecting how each type of printer physically produces output.

Metric	Full Form	Typically Used For
CPS	Characters Per Second	Old dot-matrix printers, which form output one character at a time
LPM	Lines Per Minute	High-speed industrial line printers, which print an entire line at once
PPM	Pages Per Minute	The standard speed measurement for modern laser and inkjet printers
DPI	Dots Per Inch	A measurement of print resolution/quality, not speed — higher DPI means a sharper, more detailed printed image

Remember: CPS, LPM, and PPM all measure printing *speed* (just at different granularities — character, line, or page), while DPI measures printing *quality*, not speed.

End of Study Guide — Unit 5: Emerging Technologies and Virus. Review the comparison tables and glossary above for quick revision before exams.